

A. BUSINESS REQUIREMENTS SPECIFICATION

1. INTRODUCTION

1.1 Problem

The university clinic of National University Dasmariñas serves as the main health care facility on campus and the first point of contact for anyone who needs medical attention. On an average day close to one hundred people make use of its services, including college students, senior high school students who are minors, faculty, staff, employees, and outside visitors seeking medical clearances, checkups, dental care, or first aid. Despite this steady volume, the clinic still relies almost entirely on a paper based process for recording, scheduling, and tracking every visit.

Every student health record is written by hand and stored inside locked filing cabinets arranged by academic program. Paper based systems like this often scatter patient information and make it hard for staff to see a complete medical history, and searching for files is slow and prone to error (Fatajo & Sa'ad, 2026). When a folder is misplaced, nurses often create a new record for the same student, resulting in duplicate files. This filing method also leaves out anyone who does not belong to an academic program. Since the clinic is expected to accommodate anyone within the campus vicinity, a parent, a vendor, or any other outside visitor has no existing folder and no consistent way to be recorded, leaving the clinic without a complete picture of everyone it actually serves.

Waiting time is another major concern, especially during peak periods such as freshman enrollment. The clinic mainly accepts walk in patients, and the appointment form the nurses made on Microsoft Forms does not show a queue, so students have no way to know how long the wait might be. Long waiting time remains one of the most common complaints among clinic patients everywhere (Kwak, 2023). Studies on digital appointment systems report large gains once patients can book ahead, with one hospital pharmacy system cutting average waiting time from 87 to 29 minutes (Maifo, Nlooto, & Vagiri, 2026), and an AI supported system cutting booking time by more than ninety percent while reducing manual staff intervention by seventy percent (Elhag, Althagafi, & Almouabdi, 2026). Without a similar system, the clinic leaves urgent and minor cases sitting in the same unsorted line, and the doctor and dentist schedule is posted only on a clinic Facebook page many students do not follow, so some still show up on days when no doctor is present.

The clinic staff also face a demanding challenge in managing medicine and supplies, tracked only through handwritten logbooks and an Excel file the five nurses must update by hand daily. A recent hospital pharmacy study found that manual inventory management is error prone, and that a digital inventory tool cut weekly cycle time by more than half while lowering the error rate from about nine percent to about two percent (Li, Chen, & Xiong, 2026). A wider Asia Pacific review found the same



pattern, noting that facilities without automated tracking struggle to prevent shortages and produce reliable reports (Yap, Cheong, & Yeap, 2026). This same manual counting also feeds the monthly summary report, where consultation and medicine data from the Excel file are copied by hand into a set report form, so accuracy depends entirely on every earlier entry being correct.

The current process also carries a privacy and accountability risk. Health information is sensitive personal information under Republic Act No 10173, the Data Privacy Act of 2012, yet a paper file offers no control over who can read or remove it and no record of who did so. Research on electronic health record systems shows that role based access, where each staff type only sees what their role needs, reduces unauthorized viewing and creates a traceable record of every access, something paper cannot offer (Saleh & Ebad, 2026). This gap is felt differently across the clinic. Nurses carry the heaviest load searching cabinets and updating records by hand, doctors and dentists lose time on manual paperwork and sometimes face disputes when a certificate does not match what a student expected, and minors require a parent for release while college students do not, all without any digital record tying a visit, a diagnosis, and a released document together.

These difficulties go beyond convenience. The Philippine Universal Health Care Law already calls on primary care providers to move records into electronic systems, and a pilot study on this shift found that a properly adopted digital system supports more consistent and privacy conscious care once staff are trained for the transition (Elepaño et al., 2025). The problem, then, is not the dedication of the staff who continue to serve close to a hundred people a day, but the outdated paper based process around them, one that a modern, centralized, and properly designed clinic management system is now needed to replace.



1.2 Objectives

General Objective:

The general objective of this project is to design and develop MediSync, an integrated clinic management system for the university clinic of National University Dasmariñas, that turns the current paper based process into a secure, accessible, and organized digital system, so that patient care becomes faster, health records stay protected, and clinic staff gain reliable tools for daily operations, whether the patient is an enrolled member of the university or an outside visitor within the campus vicinity.

Specific Objectives:

1. To build a web based portal for clinic staff and a mobile application for students, faculty, and staff, providing an electronic health record module that stores digital records for enrolled members through RFID tap in check in and a guided registration process for outside visitors, together with a smart appointment module that lets users book a visit, view the doctor and dentist schedule, and have that booking linked to their tap in arrival, removing paper folders and preventing duplicate records.
2. To develop a digital queue, medicine inventory, and reporting module that sorts each booked or walk in visit by urgency, keeps a live automated record of medicine stock with low stock alerts, and generates monthly reports for administrators, reducing wait times for urgent cases and removing the need for manual counting and handwritten reports.
3. To create a prescription, medical certificate, and communication module protected by role based access control and a terms and conditions consent process, that delivers notifications within the application and through the existing school Outlook email, connects to the university's existing student information system, and includes a gamified health education feature, so prescriptions and certificates are issued faster, patient data stays secure, and the university community stays informed and engaged.

1.3 Purpose

This system is being developed to directly address the problems identified in the clinic's current paper based process, and to give every person who depends on the clinic, whether a student, a staff member, or an outside visitor within the campus vicinity, a faster, safer, and more organized way to receive care.

For patients, the system is meant to remove the uncertainty of waiting in an unsorted line by allowing appointments to be booked ahead of time and each visit to be sorted by urgency, so people are attended to based on the severity of their condition rather than only the order they arrived in. A terms and conditions and consent process makes sure every patient understands how their information will be used before it is collected, and both enrolled members and outside visitors are given a clear and consistent way to be recorded.

For clinic staff, the system is meant to reduce the time spent on paperwork so more time can go toward actual patient care. Medical records become digital, complete, and easy to find, medicine stock is tracked automatically with low stock alerts, and prescriptions and certificates can be issued faster since each one is tied to an actual recorded visit. Role based access control gives each staff type, such as a nurse, doctor, or dentist, only the access their role needs, keeping patient data more secure and reducing the chance of unauthorized viewing.

For administrators, the system is meant to provide a clearer view of the clinic's day to day operations through automated monthly reports, removing the dependence on handwritten counts and giving a more accurate basis for decisions on staffing and supplies.

Beyond direct patient care, the system also aims to keep the university community informed and engaged. Notifications delivered within the application and through the existing school Outlook email are meant to make sure students and staff do not miss appointment confirmations or clinic announcements, a simple communication feature lets staff reach a student or a parent directly when needed, and a gamified health education feature gives students an easy way to learn basic health information.

Overall, this system is expected to make clinic operations faster, safer, and more reliable, reducing the workload of clinic staff, giving administrators a clearer view of daily operations, and giving every member of the university community a more convenient and dependable healthcare experience.

1.4 Definition of Terms

This section defines the key terms and technical concepts referenced throughout this documentation, to ensure a shared and consistent understanding of the terminology used across the following chapters.

Table 1. Definition of Terms

Term	Definition
Authorization	The process of determining what an authenticated user is allowed to view or do within the system based on their role.
Consent	A clear and informed agreement given by a patient or visitor allowing the clinic to collect, store, and use their personal information within the system.
Database	An organized collection of data that can be easily stored, accessed, and updated.
Encryption	A method of converting data into a coded form to protect it from being read or accessed by unauthorized users.
ERD	Short for Entity Relationship Diagram, a diagram that shows the entities within a system and the relationships between them, used to design the structure of a database.
Hashing	A process that turns input data into a unique fixed length code using a mathematical formula.
Notification	A short message sent to a user inside the application or through email to inform them of an update such as an appointment confirmation or a clinic announcement.
Query	A request made to a database to retrieve, add, update, or delete specific information.
RFID	Short for Radio Frequency Identification, a technology that lets a system identify and retrieve information linked to a card or tag once it is placed near a reader, used in this system for tap in check in.
Role-Based Access Control	A security method that limits what a user can view or edit in the system based on their assigned role, such as nurse, doctor, dentist, or administrator.
Triage	The process of sorting people to decide which ones are the most serious and must be handled first.

1.5 References

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